Short
07.01.13

L-2/T-1/CSE **Date : 07/01/2013**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2011-2012

Sub : **MATH 241** (Complex Variables and Statistics)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

Symbols used have their usual meaning.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Find all the roots of $\left(-2\sqrt{3}-2i\right)^{1/4}$. **(10)**

   (b) Show that the equation $\left|z+\frac{1}{z}\right|=2$ represents two circles with centres (0, 1), (0, –1) and radius $\sqrt{2}$. **(10)**

   (c) Prove that $u = e^{-x}\left(x\sin y - y\cos y\right)$ is harmonic. Find its harmonic conjugate v and express u + iv as an analytic function of z. **(15)**

2. (a) If f(z) is an analytic function of z prove that $\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$. **(10)**

   (b) Find the polar form of Cauchy-Riemann equations $u_x = v_y$ and $u_y = -v_x$ and hence find the Lappace's equation in polar form. **(15)**

   (c) Express sinhz in the complex form a + ib and show that $|\sinh z|^2 = \sinh^2 x + \sin^2 y$. **(10)**

3. (a) State and prove Cauchy integral formula. Use this formula to evaluate $\int_C \frac{z}{z^2+1}dz$ where C is the circle $|z + i| = 1$. **(18)**

   (b) Evaluate $\int_C \frac{z^4+1}{z^2(z+2)(2z+1)}dz$ where C is the circle $|z| = 1$. **(17)**

4. (a) State Laurent's theorem. Expand $f(z)=\frac{z}{(z-1)(2-z)}$ in a Laurent series valid for (i) $1 < |z| < 2$ (ii) $0 < |z-2| < 1$. **(18)**

   (b) Show that the transformation $w=\frac{5-4z}{4z-2}$ transforms the circle $|z| = 1$ into a circle in the w plane. Find the centre and radius of this circle. **(17)**

Contd P/2

MATH 241

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. Evaluate the following integrals by using Contour integration: **(17+18)**

(a) $\int_{-\infty}^{\infty} \frac{x \sin \pi x \, dx}{x^2 + 2x + 5}$ (b) $\int_{0}^{\infty} \frac{x^2 dx}{(x^2+9)(x^2+4)^2}$

6. (a) From the data given below calculate Pearson's first coefficient of Skewness and Kurtosis. Comment on the nature of the distribution. **(20)**

Age in years:	21 – 26	27 – 32	33 – 38	39 – 44	45 – 50	51 – 56
No. of persons:	83	75	176	130	92	54

(b) A time study was conducted in a factory with the help of two samples A and B consisting of 7 workers. The time taken by the workers in each case was recorded. From the information given below state which of the samples is more variable and which takes less time on an average. **(15)**

Time taken in minutes

Sample A:	130	124	119	134	142	130	141
Sample B:	132	144	136	143	132	133	140

7. (a) Find the two regression lines and the coefficient of correlation from the table given below: **(20)**

X:	105	112	104	114	117	108	106	117	111
Y:	58	56	47	46	51	64	66	50	55

(b) (i) An insurance company finds that the probability of dying of an individual belonging to a certain age group within a year is 0.0005. If 1000 persons of this age group buy policy of this company, find the probability that the company will receive 5 or more claims during a given year. **(7)**

(ii) The probability that A can solve a problem is $\frac{3}{7}$, that B can solve it is $\frac{2}{3}$, that C can solve it is $\frac{4}{5}$. If all of them try independently, find the probability that the problem will be solved. **(8)**

Contd P/3

MATH 241

8. (a) Explain type-I error and type-II error. 9 items of a sample had the following values:

50, 48, 49, 53, 52, 47, 45, 46, 51

Can this sample be regarded as taken from the population having 47 as mean at 5% level of significance? (Given that for $\nu = 8$, $t_{0.05} = 3.355$) **(17)**

(b) 5 coins are tossed 3200 times and the number of heads appearing each time are noted. At the end, the following results were obtained. **(18)**

No. of heads:	0	1	2	3	4	5
Frequency:	80	570	1100	900	500	50

Test the hypothesis that the coins are unbiased at 5% level of significance. (Given that for $\nu = 5$, $\chi^2_{0.05} = 11.07$)

Extra

L-2/T-1/CSE **Date : 02/06/2014**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2012-2013

Sub : **MATH 241** (Complex Variables and Statistics)

Full Marks: 210 Time : 3 Hours

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

Symbols have their usual meaning.

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE.**

1. (a) Prove the following triangle inequalities of complex numbers **(17)**

(i) $|z_1+z_2|\le|z_1|+|z_2|$ (ii) $|z_1-z_2|\ge\|z_1|-|z_2\|$.

(b) Find all the roots of the following equations in rectangular coordinates, exhibit them geometrically, and point out which is the principal root. **(18)**

(i) $z^4+2\sqrt{3}+2i=0$; (ii) $z^3+1-i=0$

2. (a) Define analytic function and harmonic function. Show that $u(x,y)=3x^2y+2x^2-y^3-2y^2$ is harmonic in some domain and find its harmonic conjugate $v(x,y)$. Express $f(z)=u+iv$ in terms of z. **(17)**

(b) Evaluate $\int_C \bar{z}dz$ from $z = 0$ to $z = 4 + 2i$ along the curve C given by (i) $z=t^2+it$;

(ii) the line from $z = 0$ to $z = 2i$ and then from $z = 2i$ to $z = 4 + 2i$. **(18)**

3. (a) (i) If $f(z)$ is integrable along a curve C having finite length L and if there exists a positive number M such that $|f(x)|\le M$ on C, then show that $\left|\int_c f(z)dz\right|\le ML$. **(13)**

(ii) Use the above result to show that $\left|\int_c \frac{dz}{z^2-1}\right|\le\frac{\pi}{3}$, where C is the contour $|z|=2$ in the first quadrant. **(5)**

(b) Use Cauchy's Integral formula to evaluate $\int_C \frac{\cos z}{z(z^2+8)}dz$; where C is the boundary of the square defined by the lines $x=\pm2$, described in the positive sense. **(17)**

4. (a) Determine the poles of $f(z)=\frac{e^z}{(z^2+\pi^2)^2}$ and find residues at each pole then use residue theorem to evaluate $\oint_C \frac{e^z}{(z^2+\pi^2)^2}dz$ where C is the circle $|z|=4$. **(18)**

(b) Find a bilinear transformation which maps the upper half of the z plane into the unit circle in the w plane in such a way that $z = i$ mapped in to $w = 0$ while the point at infinity mapped into $w=-1$. **(17)**

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE.**

5. Evaluate the following integrals by using Contour integration: **(18+17)**

(a) $\int_0^{2\pi}\frac{\cos2\theta}{5+4\cos\theta}d\theta$ (b) $\int_{-\infty}^{\infty}\frac{x^2}{(x^2+1)^2}dx$

Contd P/2

MATH 241/CSE

6. (a) Following are the runs obtained by two players A and B in 10 matches: **(15)**

Runs scored by A:	55	80	78	49	52	76	78	76	62	76
Runs scored by B:	54	78	55	66	63	69	74	64	59	85

Compute:

(i) Pearson's coefficient of skewness and kurtosis and Bowley's coefficient of skewness and comment on the shape of the runs scored by B only.

(ii) If the consistency of performance is the criterion for awarding a prize, who should get the prize?

(b) In testing a certain kind of truck tire over a rugged terrain, it is found that 20% of the trucks fail to complete the test run without a blowout. Of the next 18 trucks tested, find the probability using (both binomial and Posisson distribution) that **(10)**

(i) from 4 to 7 have blowouts;

(ii) fewer than 3 have blowouts;

(iii) more than 5 have blowouts.

(c) Derive the mean and variance of the frequency function for the binomial distribution. **(10)**

7. (a) Calculate two regression equations and regression coefficients from the data given below: **(18)**

Marks in Economics (Out of 50):	47	38	34	44	32	41	35	32	42	30
Marks in Business (Out of 50) :	38	34	42	39	34	33	30	41	38	28

Estimate the marks in economics if a student has scored 36 in business and the mark in business if another student has scored 44 in economics. Also comment on the correlation of the above mentioned marks obtained in economics and business.

(b) A survey of 640 families with 5 children each revealed the following distribution: **(17)**

No. of boys:	5	4	3	2	1	0
No. of girls:	0	1	2	3	4	5
No. of families:	28	112	220	176	80	24

Is this result consistent with the hypothesis that male and female births are equally probable? (use a 5% level of significance, given that for $\nu = 5$, $\chi^2_{0.005} = 11.07$)

8. (a) For the joint density function **(18)**

$$\begin{cases} \dfrac{6-x-y}{3}, & 0 < x < 2,\ 2 < y < 4 \\ 0, & \text{elsewhere,} \end{cases}$$

Find $P(1 < Y < 3 \mid x = 2)$.

(b) In a survey of buying habits, 500 women shoppers are chosen at random in **super market *A***. Their average weekly food expenditure is Tk. 500 with a standard deviation of Tk. 80. For another group of 500 women shoppers chosen at random in **super market B** located in another area of the same city, the average weekly food expenditure is Tk. 440 with a standard deviation of Tk. 90. Test at 10% level of significance whether the average weekly food expenditures of the populations of women shoppers are equal. **(17)**

L-2/T-1/CSE **Date : 13/12/2014**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

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Sub : **MATH 241** (Complex Variables and Statistics)

Full Marks: 210 Time : 3 Hours

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

Symbols used have their usual meaning.

1. (a) Prove that $|z_1+z_2|^2+|z_1-z_2|^2=2|z_1|^2+2|z_2|^2$ and hence deduce that $\left|a+\sqrt{a^2-b^2}\right|+\left|a-\sqrt{a^2-b^2}\right|=|a+b|+|a-b|$ where z_1, z_2, a and b are complex numbers. **(20)**

   (b) Find all of the roots of $\left(-2\sqrt{3}-2i\right)^{\frac{1}{4}}$ in rectangular coordinates and exhibit the distinct roots graphically. **(10)**

   (c) Describe mathematically and graphically the region represented by $|z+2-3i|+|z-2+3i|<10$. **(5)**

2. (a) Show that $f(z)=|z|^2$ is differentiable only at $z=0$. **(5)**

   (b)Show that $u(x,y)=\frac{1}{2}\log\left(x^2+y^2\right)$ is harmonic in some domain. Find a function $v(x,y)$ such that $f(z)=+iv$ is analytic in that domain and express $f(z)$as a function of z. **(15)**

   (c) Derive the Cauchy-Riemann equations and hence find the polar form of these equations. **(15)**

3. (a) Solve the equation $\cosh z=-2$ by equating the real and imaginary parts in the equation. **(8)**

   (b) Find the image of the hyperbola $x^2-y^2=1$ under the transformation $w=\frac{1}{z}$ and sketch the graph of the image. **(12)**

   (c) Evaluate the integral $\int_C \bar{z}dz$ around

   (i) the circle $|z-2|=3$. **(5)**

   (ii) the ellipse $|z-3|+|z+3|=10$. **(10)**

Contd P/2

MATH 241/CSE

4. (a) Evaluate $\int_C \frac{z}{(z+2)(z-4i)}dz$ by Cauchy's integral formula around the circle $C: |z-i|=4$ taken in positive sense. **(8)**

(b) Find a Laurent series expansion for $f(z)=\frac{1}{(z+1)(z-3i)}$ in power of $(z+1)$ and state the region of convergence of the series. **(8)**

(c) Find the residue of the function $f(z)=\frac{(\log z)^3}{z^2+1}$ at the point $z=i$. **(8)**

(d) Use Cauchy's residue theorem to evaluate $\int_C \frac{1+z^2}{(z-1)^2(z+2i)}dz$ where C is the circle $|z|=3$. **(11)**

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE.**

5. Evaluate the following integrals by using contour integration

(a) $\int_0^{2\pi} \frac{d\theta}{1+a\cos\theta}(-1<a<1)$ **(18)**

(b) $\int_0^{\infty} \frac{x^2}{x^6+1}dx$ **(17)**

6. (a) Lives of two models of refrigerators in a survey are as follows **(15)**

Life (No. of years)		0-2	2-4	4-6	6-8	8-10	10-12
No. of refrigerators	Model A	5	16	13	7	5	4
	Model B	2	7	12	19	9	1

Which model has greater uniformity.

(b) Calculate efficiency of skewness and kurtosis from the table given below. **(20)**

Class	10-20	20-30	30-40	40-50	50-60
Frequency	174	86	32	181	27

Also comment on the nature of the distribution.

Contd P/3

MATH 241/CSE

7. (a) Calculate the two regression equations and the coefficient of correlation from the data give below: **(20)**

Age of husband (in years)	22	25	28	31	35	32	37	39
Age of wife (in years)	19	18	22	29	31	23	30	33

Also estimate the most likely age of husband when wife's age is 24.

(b) (i) 3 cards are drawn from a deck of 52 cards one after another without replacement. Find the probability that they will come from different suits. **(7)**

(i) If 10% of the instruments produced by a company is defective, find the probability that out of 12 instruments, at least 4 instruments will be defective. **(8)**

8. (a) Talcum power is packed into tins by a machines A random samples of 11 tins is drawn and their contents are found to weigh (in lbs) as follows: **(20)**

0.44, 0.51, 0.49, 0.52, 0.45, 0.48, 0.46, 0.45, 0.47, 0.45 and 0.47

The whether the average packing can be taken to be 0.5 lbs at 5% level of significance. (Given: $t = \pm 2.23$)

(b) A survey of 320 families with 5 children each revealed the following distribution. **(15)**

No. of boys and girls	5 boys 0 girl	4 boys 1 girl	3 boys 2 girls	2 boys 3 girls	1 boys 4 girls	0 boys 5 girls
No. of families	18	56	110	88	40	8

Is the result consistent with the hypothesis that male and female births are equally probable? Use a 5% level of significance. (Given for $\nu = 5$, $\chi^2 = 11.07$)

L-2/T-1/CSE **Date : 06/02/2016**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

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Sub : **MATH 241** (Complex Variables and Statistics)

Full Marks: 210 Time : 3 Hours

USE SEPARATE SCRIPTS FOR EACH SECTION

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SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Prove that $|z_1 + z_2| \geq \big||z_1| - |z_2|\big|$ where z_1, z_2 are nonzero complex numbers. **(8)**

   (b) Show that the equation $|z - z_0| = R$ represents a circle centered at z_0 and with radius R which can be written as $|z|^2 - 2\,\mathrm{Re}\left(z\overline{z_0}\right) + |z_0|^2 = R^2$. **(7)**

   (c) Describe mathematically and graphically the region represented by $z(\bar{z} + 2) = 3$. **(10)**

   (d) Find the principal argument of $z = \left(\sqrt{3} - i\right)^6$. **(10)**

2. (a) Test the continuity and differentiability of the function $f(z)$ at the point $z = 0$, where **(10)**

$$f(z) = \begin{cases} \dfrac{(\bar{z})^2}{z}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

   (b) Find the bilinear transformation which maps the points $z_1 = 2$, $z_2 = i$ and $z_3 = -2$ into the points $w_1 = 1$, $w_2 = i$ and $w_3 = -1$, respectively. **(10)**

   (c) Derive the Cauchy-Riemann equations and hence find the polar form of these equations. **(15)**

3. (a) Solve the equation $\sin z = 2$ by equating the real and imaginary parts in the equation. **(10)**

   (b) Show that Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ continues to hold when θ is replaced by z. **(5)**

   (c) Evaluate $\int_C \left(z^2 + 3z\right)dz$ along the circle $|z| = 2$ from (2,0) to (0,2). **(10)**

   (d) Using Cauchy integral formula, find the value of the integral $\int_C \dfrac{dz}{\left(z^2 + 4\right)^2}$ around the circle $C: |z - i| = 2$, taken in the positive sense. **(10)**

Contd P/2

MATH 241

4. (a) State Laurent's theorem. Expand $f(z)=\frac{z}{(z-1)(2-z)}$ in a Laurent series valid in the region: (i) $1<|z|<2$, (ii) $0<|z-2|<1$. **(18)**

(b) State Cauchy's residue theorem and use this theorem to find the value of the integral $\int_C \frac{3z^3+2}{(z-1)(z^2+9)}dz$ taken counterclockwise around the circle (i) $|z-2|=2$ (ii) $|z|=4$. **(17)**

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. Evaluate the following integrals by using contour integration, **(17+18)**

(a) $\int_0^\infty \frac{\cos ax\,dx}{(x^2+b^2)^2}\,(a>0, b>0)$ (b) $\int_0^{2\pi} \frac{\cos 2\theta d\theta}{1-2a\cos\theta+a^2}\,(a^2<1)$

6. (a) You are given the following incomplete frequency distribution. It is known that the total frequency is 1000 and the median is 413.11. Estimate by calculation the missing frequencies and find the value of the mode. **(17)**

Class interval	Frequency
300-325	5
325-350	17
350-375	80
375-400	?
400-425	326
425-450	?
450-475	88
475-500	9

(b) Calculate coefficient of skewness by Karl Pearson's method and values of β_1 and β_2 from the following data: **(18)**

Class interval	Frequency
10-20	18
20-30	20
30-40	30
40-50	22
50-60	10

Contd P/3

MATH 241

7. (a) The following table gives the relative values of two variables. **(20)**

X:	42	44	58	55	89	98	66
Y:	56	49	53	58	65	76	51

Find the regression line of Y on X and X on Y. Find also the correlation coefficient between X and Y.

(b) A bulb manufacturing company claims that the average longevity of their bulb is 4 years with a standard deviation 0.16 years. A random sample of 10 bulbs gave mean longevity of 3.45 years. Does the sample mean justify the claim of the manufacturer? Use a 5 percent level of significance. (Necessary Chart 1 is attached) **(15)**

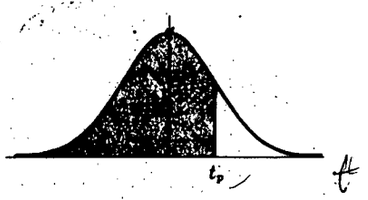
8. (a) A box contains 8 red, 3 white and 9 blue balls. If three balls are drawn at random, determine the probability that (i) all 3 are red (ii) 2 are red and 1 is white (iii) at least 1 is white. **(12)**

(b) The probability that an entering college student will graduate is 0.4. Determine the probability that out of 5 students (i) none (ii) at least 1 (iii) all will graduate. **(12)**

(c) A workshop produces 2000 units per day. The average weight of units is 130 kg. with a standard deviation of 10 kg. Assuming normal distribution, how many units are expected to weigh less than 142 kg? (Necessary Chart 2 is attached). **(11)**

Appendix III

Percentile Values (t_p) for Student's t Distribution with ν Degrees of Freedom (shaded area = p)



ν	$t_{.995}$	$t_{.99}$	$t_{.975}$	$t_{.95}$	$t_{.90}$	$t_{.80}$	$t_{.75}$	$t_{.70}$	$t_{.60}$	$t_{.55}$
1	63.66	31.82	12.71	6.31	3.08	1.376	1.000	.727	.325	.158
2	9.92	6.96	4.30	2.92	1.89	1.061	.816	.617	.289	.142
3	5.84	4.54	3.18	2.35	1.64	.978	.765	.584	.277	.137
4	4.60	3.75	2.78	2.13	1.53	.941	.741	.569	.271	.134
5	4.03	3.36	2.57	2.02	1.48	.920	.727	.559	.267	.132
6	3.71	3.14	2.45	1.94	1.44	.906	.718	.553	.265	.131
7	3.50	3.00	2.36	1.90	1.42	.896	.711	.549	.263	.130
8	3.36	2.90	2.31	1.86	1.40	.889	.706	.546	.262	.130
9	3.25	2.82	2.26	1.83	1.38	.883	.703	.543	.261	.129
10	3.17	2.76	2.23	1.81	1.37	.879	.700	.542	.260	.129
11	3.11	2.72	2.20	1.80	1.36	.876	.697	.540	.260	.129
12	3.06	2.68	2.18	1.78	1.36	.873	.695	.539	.259	.128
13	3.01	2.65	2.16	1.77	1.35	.870	.694	.538	.259	.128
14	2.98	2.62	2.14	1.76	1.34	.868	.692	.537	.258	.128
15	2.95	2.60	2.13	1.75	1.34	.866	.691	.536	.258	.128
16	2.92	2.58	2.12	1.75	1.34	.865	.690	.535	.258	.128
17	2.90	2.57	2.11	1.74	1.33	.863	.689	.534	.257	.128
18	2.88	2.55	2.10	1.73	1.33	.862	.688	.534	.257	.127
19	2.86	2.54	2.09	1.73	1.33	.861	.688	.533	.257	.127
20	2.84	2.53	2.09	1.72	1.32	.860	.687	.533	.257	.127
21	2.83	2.52	2.08	1.72	1.32	.859	.686	.532	.257	.127
22	2.82	2.51	2.07	1.72	1.32	.858	.686	.532	.256	.127
23	2.81	2.50	2.07	1.71	1.32	.858	.685	.532	.256	.127
24	2.80	2.49	2.06	1.71	1.32	.857	.685	.531	.256	.127
25	2.79	2.48	2.06	1.71	1.32	.856	.684	.531	.256	.127
26	2.78	2.48	2.06	1.71	1.32	.856	.684	.531	.256	.127
27	2.77	2.47	2.05	1.70	1.31	.855	.684	.531	.256	.127
28	2.76	2.47	2.05	1.70	1.31	.855	.683	.530	.256	.127
29	2.76	2.46	2.04	1.70	1.31	.854	.683	.530	.256	.127
30	2.75	2.46	2.04	1.70	1.31	.854	.683	.530	.256	.127
40	2.70	2.42	2.02	1.68	1.30	.851	.681	.529	.255	.126
60	2.66	2.39	2.00	1.67	1.30	.848	.679	.527	.254	.126
120	2.62	2.36	1.98	1.66	1.29	.845	.677	.526	.254	.126
∞	2.58	2.33	1.96	1.645	1.28	.842	.674	.524	.253	.126

Source: R. A. Fisher and F. Yates, *Statistical Tables for Biological, Agricultural and Medical Research* (5th edition), Table III, Oliver and Boyd Ltd., Edinburgh, by permission of the authors and publishers.

Chart 1 for Q. No 7(b)

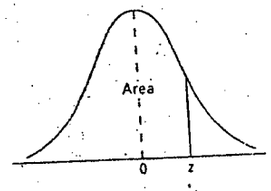


TABLE A.3 Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0352	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0722	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

TABLE A.3 (continued) Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Chart 2 for Q NO 8(c)

L-2/T-1/CSE **Date : 07/08/2016**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2015-2016

Sub : **MATH 241** (Complex Variable and Statistics)

Full Marks: 210 Time : 3 Hours

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Find all roots of $\left(-2\sqrt{3}-2i\right)^{\frac{1}{4}}$ in rectangular coordinates and exhibit the distinct roots graphically. **(13)**

   (b) Describe mathematically and graphically the region represented by $z(\bar{z}+2)=3$. **(10)**

   (c) Find the image of $\mathrm{Re}\, z \geq 0$ under the bilinear transformation $w=\frac{z-1}{z+1}$. Sketch the region $\mathrm{Re}\, z \geq 0$ an its image. **(12)**

2. (a) Prove that $u(x,y)=e^x(x\cos y - y\sin y)$ is a harmonic function. Hence find a function $v(x,y)$ such that $f(z)=u(x,y)+iv(x,y)$ is analytic. Also, express $f(z)$ as a function of z. **(15)**

   (b) Show that $Log\left(-1+\sqrt{3}i\right)^2 \neq 2Log\left(-1+\sqrt{3}i\right)$. **(10)**

   (c) Solve $\cosh z = -2$ by equating the real and imaginary parts in the equation. **(10)**

3. (a) Evaluate $\int_C \bar{z}dz$ from $z = 0$ to $z = 4 + 2i$ along the curve C given by (i) $z = t^2+it$;

   (ii) the line from z = 0 to z = 2i and then from z = 2i to z = 4 + 2i. **(16)**

   (b) State and prove Cauchy's Integral formula. **(12)**

   (c) Use Cauchy's Integral formula to evaluate $\int_C \frac{\cos z}{z\left(z^2+8\right)}dz$ **(7)**

   where C is the positively oriented circle $|z|=2$.

4. (a) What is Residue? Explain $f(z)=\frac{1}{(z+1)(z+3)}$ in a Laurent's series valid for **(17)**

   (i) $|z|>3$, (ii) $0<|z+1|<2$. Write down the residue in each case.

   (b) Evaluate $\frac{1}{2\pi i}\int_C \frac{e^{zt}}{z^2\left(z^2+2z+2\right)}dz$ around the circle C with equation $|z|=3$. **(18)**

Contd P/2

MATH 241 (CSE)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. Evaluate the following integrals using residues and contours:

(a) $\int_{-\infty}^{+\infty} \frac{x^2}{(x^2+1)^2(x^2+2x+2)}dx$ **(17)**

(b) $\int_{0}^{\infty} \frac{x^3 \sin x}{(x^2+1)(x^2+9)}dx$ **(18)**

6. (a) For the following incomplete frequency distribution, it is known that the total frequency is 1000 and the median is 415. Estimate the missing frequencies and hence find the modal value of the completed table: **(18)**

Variable	Frequency
300 – 325	5
325 – 350	17
350 – 375	80
375 – 400	?
400 – 425	326
425 – 450	?
450 – 475	88
475 – 500	9

Also, comment on the shape of the frequency distribution.

(b) During the 10 weeks of a session, marks scored by two students, X and Y, in a subject are given below: **(17)**

X	58	59	60	54	65	66	52	75	69	52
Y	87	89	78	71	73	84	65	66	56	46

(i) Who is better scorer?

(ii) Who is more consistent?

7. (a) Calculate coefficient of skewness and kurtosis from the table given below: **(18)**

Class	10–20	20–30	30–40	40–50	50–60
Frequency	174	86	32	181	27

Also, comment on the nature of the distribution.

Contd P/3

MATH 241 (CSE)
Contd... Q. No. 7

(b) A factory manager wants to find a measure which he can use to fix the monthly income for persons applying for a job in a production department. As an experimental project, he collected the following data of 9 persons from that department referring to years of service and their monthly income. **(17)**

Year of Service	10	8	9	5	7	6	12	11	13
Income in '000	17	12	14	8	10	9	20	18	22

Calculate the following:

(i) The regression equations and the correlation coefficient.

(ii) What initial start would you recommend for a person applying for a job after having served in a similar capacity in another factory for 15 years?

8. (a) The probability that a married man watches a certain television show is 0.25 and that a married woman watches the show is 0.35. The probability that a man watches the show, given that his wife does, is 0.8. Find the probability that **(13)**

(i) a married couple watches the show.

(ii) a husband watches the show given that his wife does not.

(b) The mark in a test is assumed to be normally distributed. It is known that 10% of the students have marks under 65 and 25% exceed 75. What percentage of students have marks between 60 and 74? (Necessary chart 1 is attached) **(12)**

(c) The mean weekly sale of a chocolate bar company in candy stores was 153.7 bars per store. After an advertising campaign, the mean weekly sale in 29 stores for a typical week increased to 169.4 and showed a standard deviation of 19.7. Was the advertising successful? Use a 5% level of significance. **(10)**

(Necessary chart 2 is attached)

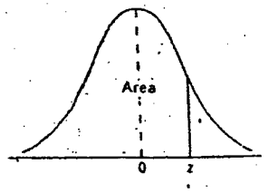


TABLE A.3 Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0352	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0722	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

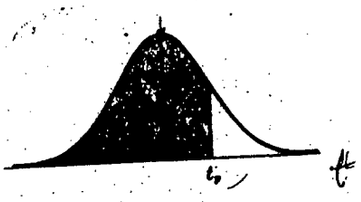
TABLE A.3 (continued) Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Chart 1 for Q. No. 8(b)

Appendix III

Percentile Values (t_p) for Student's t Distribution with ν Degrees of Freedom (shaded area = p)



ν	$t_{.995}$	$t_{.99}$	$t_{.975}$	$t_{.95}$	$t_{.90}$	$t_{.80}$	$t_{.75}$	$t_{.70}$	$t_{.60}$	$t_{.55}$
1	63.66	31.82	12.71	6.31	3.08	1.376	1.000	.727	.325	.158
2	9.92	6.96	4.30	2.92	1.89	1.061	.816	.617	.289	.142
3	5.84	4.54	3.18	2.35	1.64	.978	.765	.584	.277	.137
4	4.60	3.75	2.78	2.13	1.53	.941	.741	.569	.271	.134
5	4.03	3.36	2.57	2.02	1.48	.920	.727	.559	.267	.132
6	3.71	3.14	2.45	1.94	1.44	.906	.718	.553	.265	.131
7	3.50	3.00	2.36	1.90	1.42	.896	.711	.549	.263	.130
8	3.36	2.90	2.31	1.86	1.40	.889	.706	.546	.262	.130
9	3.25	2.82	2.26	1.83	1.38	.883	.703	.543	.261	.129
10	3.17	2.76	2.23	1.81	1.37	.879	.700	.542	.260	.129
11	3.11	2.72	2.20	1.80	1.36	.876	.697	.540	.260	.129
12	3.06	2.68	2.18	1.78	1.36	.873	.695	.539	.259	.128
13	3.01	2.65	2.16	1.77	1.35	.870	.694	.538	.259	.128
14	2.98	2.62	2.14	1.76	1.34	.868	.692	.537	.258	.128
15	2.95	2.60	2.13	1.75	1.34	.866	.691	.536	.258	.128
16	2.92	2.58	2.12	1.75	1.34	.865	.690	.535	.258	.128
17	2.90	2.57	2.11	1.74	1.33	.863	.689	.534	.257	.128
18	2.88	2.55	2.10	1.73	1.33	.862	.688	.534	.257	.127
19	2.86	2.54	2.09	1.73	1.33	.861	.688	.533	.257	.127
20	2.84	2.53	2.09	1.72	1.32	.860	.687	.533	.257	.127
21	2.83	2.52	2.08	1.72	1.32	.859	.686	.532	.257	.127
22	2.82	2.51	2.07	1.72	1.32	.858	.686	.532	.256	.127
23	2.81	2.50	2.07	1.71	1.32	.858	.685	.532	.256	.127
24	2.80	2.49	2.06	1.71	1.32	.857	.685	.531	.256	.127
25	2.79	2.48	2.06	1.71	1.32	.856	.684	.531	.256	.127
26	2.78	2.48	2.06	1.71	1.32	.856	.684	.531	.256	.127
27	2.77	2.47	2.05	1.70	1.31	.855	.684	.531	.256	.127
28	2.76	2.47	2.05	1.70	1.31	.855	.683	.530	.256	.127
29	2.76	2.46	2.04	1.70	1.31	.854	.683	.530	.256	.127
30	2.75	2.46	2.04	1.70	1.31	.854	.683	.530	.256	.127
40	2.70	2.42	2.02	1.68	1.30	.851	.681	.529	.255	.126
60	2.66	2.39	2.00	1.67	1.30	.848	.679	.527	.254	.126
120	2.62	2.36	1.98	1.66	1.29	.845	.677	.526	.254	.126
∞	2.58	2.33	1.96	1.645	1.28	.842	.674	.524	.253	.126

Source: R. A. Fisher and F. Yates, *Statistical Tables for Biological, Agricultural and Medical Research* (5th edition), Table III. Oliver and Boyd Ltd., Edinburgh, by permission of the authors and publishers.

Chart 2 for Q. No. 8(c)

L-2/T-1/CSE **Date: 05/08/2017**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2016-2017

Sub: **MATH 245** (Complex Variable and Statistics)

Full Marks: 210 Time: 3 Hours

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks

Symbols used have their usual meaning.

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Let $f(z)=u+iv$ be differentiable at a nonzero point $z_0=r_0e^{i\theta_0}$. Show that $f'(z_0)=e^{-i\theta}(u_r+iv_r)=-\frac{i}{z_0}(u_\theta+iv_\theta)$ where $u_r, v_r, u_\theta, v_\theta$ are evaluated at (r_0,θ_0). **(18)**

(b) Find the bilinear transformation which maps the points $z_1=-i, z_2=0$ and $z_3=i$ into $w_1=-1, w_2=i$ and $w_3=1$, respectively. **(12)**

(c) Show that Euler's formula $e^{i\theta}=\cos\theta+i\sin\theta$ continues to hold when θ is replaced by z. **(5)**

2. (a) Show that $Log\left(-1+i\sqrt{3}\right)^2\neq 2Log\left(-1+i\sqrt{3}\right)$. **(8)**

(b) Solve $\cosh z=-2$ by equating the real and imaginary parts in the equation. **(10)**

(c) Show that $u(x,y)=\frac{1}{2}\log\left(x^2+y^2\right)$ is harmonic in some domain. Hence find a function $v(x,y)$ such that $f(z)=u+iv$ is analytic in that domain. Also, express $f(z)$ in terms of z. **(17)**

3. (a) Use Cauchy's integral formula to evaluate the integral $\int_C \frac{z}{(9-z^2)(z+i)}dz$, where C is the circle $|z|=2$, taken in the positive sense. **(12)**

(b) Expand $f(z)=\frac{z+1}{z-2}$ in a Taylor series in powers of z and state the region where the expansion is valid. **(8)**

(c) State Laurent's theorem. Find two Laurent series of $f(z)=\frac{1}{(z+1)(z+3)}$ valid in the regions (i) $1<|z|<3$ and (ii) $0<|z+1|<2$. **(15)**

4. (a) Evaluate the integral $\int_C \bar{z}\,dz$ around **(12)**

(i) the circle $|z-2|=3$, and (ii) the ellipse $|z-3|+|z+3|=10$.

(b) Evaluate the following integrals by Cauchy's residue theorem where C is the circle $|z|=3$ taken in the counterclockwise sense: **(23)**

(i) $\int_C \frac{1+z^2}{(z-1)^2(z+2i)}dz$, (ii) $\int_C \frac{\sin z}{z^2\left(z^2+4\right)}dz$ (iii) $\int_C \frac{\sinh z}{z^4}dz$

Contd P/2

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. Evaluate the following integrals using residues and contours: **(17+18)**

(a) $\int_0^{\infty} \frac{\cos(3\theta)}{5+4\cos\theta} d\theta$ (b) $\int_0^{\infty} \frac{dx}{(a^2+x^2)^2}$

6. (a) Define Skewness and Kurtosis with classifications. Using moments calculate the coefficients of Skewness and Kurtosis from the following distribution given below and comment on the result obtained. **(17)**

Profits (in Taka)	10-20	20-30	30-40	40-50	50-60
No. of companies	18	20	30	22	10

(b) The following table shows the experience (X) and the performance rating (Y) of 5 persons: **(18)**

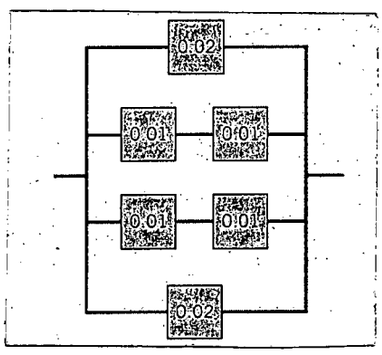
X	16	12	18	4	3
Y	87	88	89	68	58

(i) Fit a linear regression model Y on X.

(ii) Find sum of squares for regression (SSR) and total sum of squares (SST).

(iii) Is the linear model appropriate for given data? Justify your result using coefficient of determination.

7. (a) The following circuit operates if and only if there is a path of functional devices from left to right. Assume that devices fail independently and that the probability of failure of each device is as shown below. What is the probability that the circuit does not operate? **(10)**



MATH 245/CSE
Contd... Q. No. 7

(b) A U-shaped component is to be formed from the three parts A, B, and C. The picture is shown below. The length of A is normally distributed with a mean of 10 millimeters and a standard deviation of 0.1 millimeter. The thickness of parts B and C is normally distributed with a mean of 2 millimeters and a standard deviation of 0.05 millimeter. Assume all dimensions are independent. **(10)**

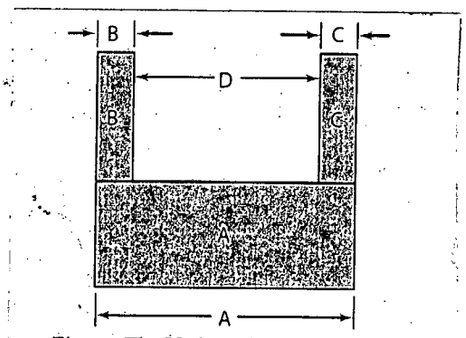


Figure: The U-shaped component.

(i) Determine the mean and standard deviation of the length of the gap D.

(ii) What is the probability that the gap D is less than 5.9 millimeters? (Necessary table attached).

(c) The masses of 1500 ball bearings are normally distributed, with a mean of 22.40 g and a standard deviation of 0.048 g. If 300 random samples of size 36 are drawn from this population, determine the expected mean and standard deviation of the sampling distribution of means if the sampling is done with replacement and without replacement. How many of the random samples would have their means (i) between 22.39 and 22.41 g, (ii) greater than 22.42 g and (iii) less than 22.38 g or more than 22.41 g? (Necessary table attached). **(15)**

8. (a) If X is a binomial random variable with probability distribution $b(x;n,p)$. When $n \to \infty$, $p \to 0$ and $np \to \mu$ remains constant then prove that $b(x;n,p) \to p(x,\mu)$. In a certain industrial facility accidents occur infrequently. It is known that the probability of an accident on any given day is 0.005 and accidents are independent of each other. What is the probability that **(20)**

(i) in any given period of 400 days there will be an accident on one day and

(ii) There are at most three days with an accident.

(b) A random sample of 100 recorded deaths in Bangladesh during the past year showed an average life span of 61.8 years. Assuming a population standard deviation of 4.9 years, does this seem to indicate that the mean life span today is greater that 60 years? Use a 0.05 level of significance. (Necessary table attached) **(15)**

=4=

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Table for question no (7 and 8)

L-2/T-1/CSE **Date: 03/11/2019**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2018-2019

Sub: **MATH 245** (Complex Variables and Statistics)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

Symbols used have their usual meaning.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) If $|z_1|=|z_2|$ and $\text{amp}(z_1)+\text{amp}(z_2)=0$, then show that z_1 and z_2 are two complex conjugate numbers. **(10)**

(b) Prove that if the ratio $\frac{z-i}{z-1}$ is purely imaginary, then the point z lies on the circle whose centre $\frac{1}{2}(1+i)$ and radius $\frac{1}{\sqrt{2}}$. **(10)**

(c) Show that the transformation, $w=\frac{1+iz}{z+i}$ transform the real axis on the z-plane onto a circle in the w-plane. Find the centre and radius of this circle. **(15)**

2. (a) Write down Cauchy Riemann equations in polar form. If $f(z)=u(r,\theta)+iv(r,\theta)$ then test the differentiability of the function $f(z)=\sqrt{r}e^{i\frac{\theta}{2}}, (r>0, \alpha<\theta<\alpha+2\pi)$ in the indicated domain and hence show that $f'(z)=\frac{1}{2f(z)}$. **(15)**

(b) Show that $u=e^x\cos y$ is a harmonic function. Find an analytic function $f(z)=u(x,y)+iv(x,y)$ and express $f(z)$ in terms of z. **(10)**

(c) Show that the integral of $\bar{z}$ along a semi-circular arc $|z|=1$ from -1 to 1 has the value $-\pi i$ or πi according as the arc lies above or below the real axis. **(10)**

3. (a) Use Cauchy integral formula to evaluate $\oint_C \frac{\sin^6 z}{\left(z-\frac{\pi}{6}\right)^3}dz$, where C is the circle $|z|=1$. **(10)**

(b) Express $f(z)=\frac{1}{(z^2+1)(z+2)}$ in a Laurent series valid in the region $1<|z|<2$. **(13)**

(c) Evaluate $\int_C \frac{1}{z(2z-1)(z-4)}dz$, where $C=\{z:|z+2|+|z-2|=6, \text{positively oriented}\}$ by Cauchy's residue theorem. **(12)**

4. Evaluate the following integral using the method of contour integration: **(17+18)**

(a) $\int_0^{2\pi}\frac{\sin 2\theta d\theta}{5-3\cos\theta}$ (b) $\int_0^{\infty}\frac{dx}{(1+x^2)^2}$.

Contd P/2

MATH 245/CSE

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) The Shareholder Research Bureau of BD conducted recently a research study on the price behavior of three leading industrial shares A, B and C for the period 1996 to 1999, the results of which are published as follows in the quarterly journal: **(15)**

Share	Average price (tk)	Standard Deviation (tk)	Current selling price (tk)
A	18.00	5.40	36.00
B	22.50	4.50	34.75
C	24.00	6.00	39.00

(i) Which share in your opinion appears to be more stable in value?

(ii) If you are the holder of all three shares which one would you like to dispose of at present and why?

(b) What is Skewness and Kurtosis? Calculate the value of γ_1 and γ_2 from the data and interpret them. **(20)**

Class interval	10-20	20-30	30-40	40-50	50-60
Frequency	18	20	30	22	10

6. (a) Find the regression equation of capacity utilisation on production from the data where $r = 0.62$ **(20)**

	Average	Standard deviation
Production	35.6	10.5
Capacity utilisation	84.8	8.5

Also estimate the production when the capacity utilisation is 70 percent.

(b) A group of 20 students went on a picnic. Among them 5 got sun burnt, 8 got bitten by mosquitoes and 10 returned home safely without any mishap. What is the probability that (i) a boy, who got sun burnt was ignored by the mosquitoes? and (ii) a mosquito bitten boy also got sun burnt? **(15)**

7. (a) What is density function? What condition must a density function satisfy? Suppose that the distribution function of a random variable X is as follows **(20)**

$$F(x)=\begin{cases}0, & x\le 0\\ \frac{x}{8}, & 0<x<2\\ \frac{y^2}{16}, & 2\le x<4\\ 1, & x\ge 4\end{cases}$$

(i) Find the probability density function of X.

(ii) Find $p(x\ge 1.5)$ and $p(x\ge 1\,|\,x\le 3)$.

Contd P/3

MATH 245/CSE

Contd... Q. No. 7

(b) A lot containing 7 components is sampled by a quality inspector, the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample taken by the inspector? **(15)**

8. (a) A new factory installed 1000 tube lights with a mean life of 120 days and their length of life is normally distributed with standard deviation 20 days. (i) How many tube lights will expire in less than 90 days? (ii) If it is decided to replace all the tube lights together, what interval should be allowed between replacements if not more than 10 percent should expire before replacement? **(18)**

(b) The mean and standard deviation of GPA scores obtained from a random sample of 40 students of a college were 2.8 and 0.35 respectively. Can we conclude that the sample has come from the entire group of students, which has a mean score of 2.4? Use a 1% level of significance. Compute 99% confidence interval for the mean score in the population. (Given that, at 1% level of significance; $z = \pm 2.58$.) **(17)**

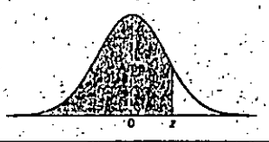


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

For Question No. 8(a)

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

For Question No. 8(a)

L-2/T-1/CSE **Date: 03/11/2019**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2018-2019

Sub: **MATH 245** (Complex Variables and Statistics)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

Symbols used have their usual meaning.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) If $|z_1|=|z_2|$ and $\text{amp}(z_1)+\text{amp}(z_2)=0$, then show that z_1 and z_2 are two complex conjugate numbers. **(10)**

(b) Prove that if the ratio $\frac{z-i}{z-1}$ is purely imaginary, then the point z lies on the circle whose centre $\frac{1}{2}(1+i)$ and radius $\frac{1}{\sqrt{2}}$. **(10)**

(c) Show that the transformation, $w=\frac{1+iz}{z+i}$ transform the real axis on the z-plane onto a circle in the w-plane. Find the centre and radius of this circle. **(15)**

2. (a) Write down Cauchy Riemann equations in polar form. If $f(z)=u(r,\theta)+iv(r,\theta)$ then test the differentiability of the function $f(z)=\sqrt{r}e^{i\frac{\theta}{2}}, (r>0, \alpha<\theta<\alpha+2\pi)$ in the indicated domain and hence show that $f'(z)=\frac{1}{2f(z)}$. **(15)**

(b) Show that $u=e^x\cos y$ is a harmonic function. Find an analytic function $f(z)=u(x,y)+iv(x,y)$ and express $f(z)$ in terms of z. **(10)**

(c) Show that the integral of $\bar{z}$ along a semi-circular arc $|z|=1$ from -1 to 1 has the value $-\pi i$ or πi according as the arc lies above or below the real axis. **(10)**

3. (a) Use Cauchy integral formula to evaluate $\oint_C \frac{\sin^6 z}{\left(z-\frac{\pi}{6}\right)^3}dz$, where C is the circle $|z|=1$. **(10)**

(b) Express $f(z)=\frac{1}{(z^2+1)(z+2)}$ in a Laurent series valid in the region $1<|z|<2$. **(13)**

(c) Evaluate $\int_C \frac{1}{z(2z-1)(z-4)}dz$, where $C=\{z:|z+2|+|z-2|=6, \text{positively oriented}\}$ by Cauchy's residue theorem. **(12)**

4. Evaluate the following integral using the method of contour integration: **(17+18)**

(a) $\int_0^{2\pi}\frac{\sin 2\theta d\theta}{5-3\cos\theta}$ (b) $\int_0^{\infty}\frac{dx}{(1+x^2)^2}$.

Contd P/2

MATH 245/CSE

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) The Shareholder Research Bureau of BD conducted recently a research study on the price behavior of three leading industrial shares A, B and C for the period 1996 to 1999, the results of which are published as follows in the quarterly journal: **(15)**

Share	Average price (tk)	Standard Deviation (tk)	Current selling price (tk)
A	18.00	5.40	36.00
B	22.50	4.50	34.75
C	24.00	6.00	39.00

(i) Which share in your opinion appears to be more stable in value?

(ii) If you are the holder of all three shares which one would you like to dispose of at present and why?

(b) What is Skewness and Kurtosis? Calculate the value of γ_1 and γ_2 from the data and interpret them. **(20)**

Class interval	10-20	20-30	30-40	40-50	50-60
Frequency	18	20	30	22	10

6. (a) Find the regression equation of capacity utilisation on production from the data where $r = 0.62$ **(20)**

	Average	Standard deviation
Production	35.6	10.5
Capacity utilisation	84.8	8.5

Also estimate the production when the capacity utilisation is 70 percent.

(b) A group of 20 students went on a picnic. Among them 5 got sun burnt, 8 got bitten by mosquitoes and 10 returned home safely without any mishap. What is the probability that (i) a boy, who got sun burnt was ignored by the mosquitoes? and (ii) a mosquito bitten boy also got sun burnt? **(15)**

7. (a) What is density function? What condition must a density function satisfy? Suppose that the distribution function of a random variable X is as follows **(20)**

$$F(x)=\begin{cases}0, & x\le 0\\ \frac{x}{8}, & 0<x<2\\ \frac{y^2}{16}, & 2\le x<4\\ 1, & x\ge 4\end{cases}$$

(i) Find the probability density function of X.

(ii) Find $p(x\ge 1.5)$ and $p(x\ge 1\,|\,x\le 3)$.

Contd P/3

MATH 245/CSE

Contd... Q. No. 7

(b) A lot containing 7 components is sampled by a quality inspector, the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample taken by the inspector? **(15)**

8. (a) A new factory installed 1000 tube lights with a mean life of 120 days and their length of life is normally distributed with standard deviation 20 days. (i) How many tube lights will expire in less than 90 days? (ii) If it is decided to replace all the tube lights together, what interval should be allowed between replacements if not more than 10 percent should expire before replacement? **(18)**

(b) The mean and standard deviation of GPA scores obtained from a random sample of 40 students of a college were 2.8 and 0.35 respectively. Can we conclude that the sample has come from the entire group of students, which has a mean score of 2.4? Use a 1% level of significance. Compute 99% confidence interval for the mean score in the population. (Given that, at 1% level of significance; $z = \pm 2.58$.) **(17)**

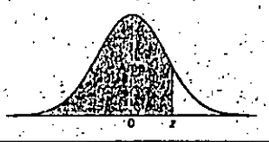


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

For Question No. 8(a)

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

For Question No. 8(a)

L-2/T-1/CSE Date: 25/09/2021

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2019-2020

Sub: **MATH 245** (Complex Variable and Statistics)

Full Marks: 210 Time: 2 Hours 30 minutes

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

SECTION - A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Show that $|\cos(z)|^2 = (\cos x)^2 + (\sinh y)^2$ for all complex numbers $z = x + iy$. (10)

   (b) Determine which of the following functions $f(z)$ are entire and which are not? Also, find the complex derivative $f'(z)$ of $f(z)$ if $f(z)$ is entire. (10)

   (i) $f(z) = \frac{1}{1+|z-1|^2}$, (ii) $f(z) = 5^{(2^z)}$

   (c) Let R be the region in the z-plane bounded by $x = 0$, $y = 0$, $x = 2$, $y = 1$. Determine the region R' of the w-plane into which R is mapped under the transformation $w = \sqrt{2}\left(e^{i\frac{\pi}{4}}\right)z + (1 - 2i)$. (15)

2. (a) Show that $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is a harmonic function. Hence find a function $v(x, y)$ such that $f(z) = u(x, y) + iv(x, y)$ is analytic. Also, express $f(z)$ as a function of z. (20)

   (b) Find the Laurent series of $f(z) = \frac{z^2}{z^2-3z+2}$ in each of the following domain: (i) $1 < |z| < 2$ and (ii) $1 < |z - 3| < 2$. (15)

3. (a) Evaluate $\int_i^{2+5i}(3x - y)dx + (2y + x)dy$ along (18)
   (i) the curve $y = x^2 + 1$,
   (ii) the straight lines from (0,1) to (2,1) and then from (2,1) to (2,5).

   (b) Evaluate the integral $\oint_C \frac{e^{-iz}}{(z+3)(z-i)^2}dz$ by Cauchy's Residue theorem, where $C = \{z: z = 1 + 2e^{i\theta}, 0 \le \theta \le 2\pi\}$. (17)

4. (a) If C is the boundary of the triangle with vertices 0, $3i$ and -4 oriented in the counterclockwise direction then show that $\left|\oint_C (e^z - \bar{z})dz\right| \le 57$. (18)

   (b) Evaluate the integral $\int_0^{\pi} \frac{d\theta}{2-\cos\theta}$ by the method of contour integration. (17)

SECTION - B

There are FOUR questions in this section. Answer any THREE.

5. (a) Calculate the coefficient of variance for the following data: (10)

Profits	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of Companies	4	8	18	30	15	10	8	7

(b) An analysis of workers resulted in the following distribution:

Earnings (Tk.)	50-70	70-90	90-110	110-130	130-150	150-170	170-190
No. of employees	4	8	12	20	6	7	3

(25)

Calculate the first four moments about assumed mean. Convert the result into moments about the mean. Calculate values of γ_1 and γ_2.

6. (a) The following table shows the distance to transmitter (X) and corresponding wireless signal strength (Y): (17)

Distance to transmitter (m)	13	12	17	19	14	15	15	8	13	3
wireless signal strength (dB)	34.4	38.4	30.4	29.7	30.1	33.9	32.8	35.2	34.9	36.8

Find the regression line of Y on X and also predict what the signal strength would be if the distance is 10 meters.

(b) In a research report, a student of Medical College claims that mice with an average life span of 32 months will live to be about 40 months old when 40% of the calories in their diet are replaced by vitamins and protein. Is there any reason to believe that $\mu < 40$ if 64 mice that are placed on this diet have an average life of 38 months with a standard deviation of 5.8 months? Use a P-value in your conclusion. (18)

7. (a) A truth serum has the property that 90% of the guilty suspects are properly judged while, of course, 10% of the guilty suspects are improperly found innocent. On the other hand, innocent suspects are misjudged 1% of the time. If the suspect was selected from a group of suspects of which only 5% have ever committed a crime, and the serum indicates that he is guilty, what is the probability that he is innocent? (17)

(b) A continuous random variable X has the density function $f(x) = \begin{cases} e^{-x}, x > 0 \\ 0, \text{ Otherwise} \end{cases}$. (18)

Find the expected value of $g(X) = e^{2X/3}$.

8. (a) In a manufacturing process where glass products are made, defects or bubbles occur, occasionally rendering the piece undesirable for marketing. It is known that, on average, 1 in every 1000 of these items produced has one or more bubbles. What is the probability that a random sample of 8000 will yield fewer than 7 items possessing bubbles? (10)

(b) Suppose that the time between emergency calls to a small suburban fire station follows an exponential distribution with an average rate of 1.8 calls per day. (25)

(i) Phil the fireman has just clocked on. What is the chance of a call in the next 15 minutes?

(ii) Phil has nearly finished his shift: 15 minutes to go. There has been no call during his shift so far. What is the chance of a call in the next 15 minutes?

(iii) Judy works a 10-hour shift, Mondays to Thursdays. What is the probability that she has no calls in a shift?

(iv) What is the probability that she has no calls in four successive days?

(v) Judy is talking about her job: 'In 10% of shifts, there's a call in the first x hours of the shift.' What is x, to one decimal place?

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2020-2021

Sub: **MATH 245** (Complex Variable and Statistics)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

Symbols used have their usual meaning.

1. (a) Test the differentiability of the function $f(z)=\frac{64}{z^4}$ in its natural domain. **(11)**

(b) Transform the circle $x^2+y^2-25x=0$ into a straight line applying the $w=\frac{5z+4}{z+3}$ transformation. **(12)**

(c) Find all roots of the equation $\cos z = 4$ by equating the real and imaginary parts in the equation. **(12)**

2. (a) Prove that $u(x,y)=\sinh x \sin y+\frac{y}{x^2+y^2}$ is harmonic. Find a harmonic conjugate of $u(x,y)$. **(15)**

(b) Let C be the arc of the circle $|z|=2$ from $z=2$ to $z=2i$ that lies in the first quadrant, then show that $\left|\oint_C \frac{z+4}{z^3-1}dz\right| \le \frac{6\pi}{7}$. **(10)**

(c) Show that $\text{Log}(-1+i)^2 \neq 2\log(-1+i)$. **(10)**

3. (a) Use Cauchy's integral formula to evaluate the integral $\frac{1}{2\pi i}\oint_C \frac{e^{-3z}}{(z^2+4)}dz$, where C is the circle $|z|=3$, taken in the positive sense. **(10)**

(b) Expand $f(z)=\frac{5}{(z+1)(z+3)}$ in a Laurent's series valid for (i) $1<|z|<3$ and (ii) $|z|<1$. **(10)**

(c) Evaluate $\oint_C \frac{\sin \pi z}{(z-3)^3}dz$, where C is the circle $|z|=4$, by Cauchy's residue theorem. **(15)**

4. Evaluate the following integral using the method of contour integration:

(i) $\int_0^{2\pi} \frac{d\theta}{(5-3\cos\theta)^2}$ **(17)**

(ii) $\int_0^{\infty} \frac{x^2 dx}{(1+x^6)}$ **(18)**

Contd P/2

MATH 245/CSE

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

Symbols used have their usual meaning.

5. (a) A survey, data on daily wages paid to workers of two factories A and B are as follows: **(10)**

Daily Wages	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Factory A	15	30	44	60	30	14	7
Factory A	25	40	60	35	20	15	5

Find out:

(i) Which factory pays higher average wages?

(ii) Which factory has greater variability about paying wages?

(b) An analysis of weight distribution resulted in the following table: **(15)**

Weight	30-40	40-50	50-60	60-70	70-80	80-90
Frequency	18	37	45	27	15	8

Calculate Bowley's coefficient of skewness and Kelly's coefficient of skewness for the above data. Hence, interpret your results by discussing the advantages and drawbacks of the methods.

(c) Each rear tire on an experimental airplane is supposed to be filled to a pressure of 40 pounds per square inch (psi). Let X denote the actual air pressure for the right tire and Y denote the actual air pressure for the left tire. Suppose that X and Y are random variables with the joint density function: **(10)**

$$f(x,y)=\begin{cases} k(x^2+y^2), & 30 \le x < 50,\ 30 \le y < 50, \\ 0, & \textit{elsewhere.} \end{cases}$$

(i) Find the value of k.

(ii) Find $P(30 \le X \le 40 \text{ and } 40 \le Y < 50)$.

(iii) Find the probability that both tires are underfilled.

6. (a) The following table shows the experience (X) and the performance rating (Y) of 5 persons: **(15)**

X	16	12	18	4	3
Y	87	88	89	68	58

(i) Fit a linear regression model for Y on X.

(ii) Find sum of squares for regression (SSR) and total sum of squares (SST).

(b) Measurement of scientific system are always subject to variation, some more than others. There are many structures for measurement error, and statisticians spend a great deal of time modeling these errors. Suppose the measurement error X of a certain physical quantity is decided by the density function: **(10)**

$$f(x) = \begin{cases} k(3-x)^2, & -1 \le x \le 1, \\ 0 & elsewhere. \end{cases}$$

(i) Determine k that renders $f(x)$ a valid density function.

(ii) Find the probability that a random error in measurement is less than $\frac{1}{2}$.

(iii) For this particular measurement. it is undesirable if the magnitude of the error exceeds 0.8. What is the probability that this occurs?

(c) According to a study published by a group of University of Massachusetts sociologists, about two-thirds of the 20 million persons in this country who take Valium are women. Assuming this figure to be a valid estimate, find the probability that on a given day the fifth prescription written by a doctor for Valium is **(10)**

(i) the first prescribing Valium for a women

(ii) the third prescribing Valium for a women.

7. (a) Imperfections in computer circuit boards and computer chips lend themselves to statistical treatment. For a particular type of board, the probability of a diode failure is 0.03 and the board contains 200 diodes. **(17)**

(i) What is the mean number of failures among the diodes?

(ii) What is the variance?

(iii) The board will work if there are no defective diodes. What is probability that a board will work?

(b) A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live **(18)**

(i) more than 32 months

(ii) between 37 and 49 months.

(Necessary Table 1 and 2 are attached)

8. (a) The hospitalization period, in days, for patients following treatment for a certain type of kidney disorder is a random variable Y = X + 4, where X has the density function: **(10)**

$$f(x) = \begin{cases} \dfrac{32}{(x+4)^3}, & x > 0. \\ 0, & elsewhere. \end{cases}$$

Find the average number of days that a person is hospitalized following treatment for this disorder.

MATH 245/CSE

Contd... Q. No. 8

(b) According to a dietary study, high sodium intake may be related to ulcers, stomach cancer, and migraine headaches. The human requirement for salt is only 220 milligrams per day, which is surpassed in most single servings of ready-to-eat cereals. If a random sample of 20 similar servings of a certain cereal has a mean sodium content of 244 milligrams and a standard deviation of 24.5 milligrams, does this suggest at the 0.05 level of significance that the average sodium content for a single serving of such cereal is greater than 220 milligrams? **(15)**

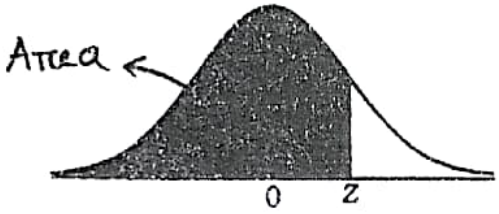
(Necessary Table 3 is attached)

(c) In a scientific paper, one researcher described the reliability of DNA paternity testing as follows: "To get a completely accurate result, you would have to be tested, and so would (the man) and your mother. The test is 100% accurate if the man is not the father and 99.9% accurate if he is." Consider using the results of DNA paternity testing to decide between the following two hypotheses: **(10)**

H_0 : a particular man is the father

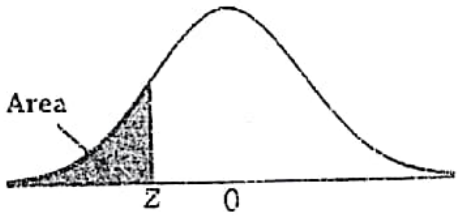
H_a : a particular man is not the father

In the context of this problem, describe Type I and Type II errors.



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Table 1: Necessary Table for Question No. 7(b)

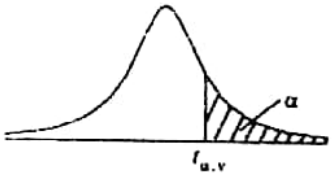


z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table 2: Necessary Table for Question No. 7(b)

Table of the Student's *t*-distribution

The table gives the values of $t_{\alpha,\nu}$ where
$\Pr(T_\nu > t_{\alpha,\nu}) = \alpha$, with ν degrees of freedom



ν \ α	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.076	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Table: 3 Necessary Table for Question No. 8(b)

L-2/T-1/CSE **Date: 30/04/2023**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2021-2022

Sub: **MATH 245** (Complex Variable and Statistics)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

Symbols used have their usual meaning.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) If $|z_1 - z_2|$ denotes the distance between two points z_1 and z_2, then identify the following and give a geometrical interpretation for: **(15)**

   (i) $|z-3i|+|z+3i|=8$

   (ii) $|z-1|=|z+i|$.

   (b) Find all the roots of $(-8)^{\frac{1}{6}}$ in rectangular coordinates, exhibit them as vertices of certain regular polygon and identify the principal root. **(10)**

   (c) Evaluate: $\lim_{z\to i}\left[\dfrac{\tan^{-1}(z^2+1)^2}{\sin^2(z^2+1)}\right]$. **(10)**

2. (a) Define harmonic function and analytic function. Prove that $u=\cos x\cosh y$ is harmonic. Also find its harmonic conjugate and hence find the corresponding analytic function $f(z)$ in terms of z. **(15)**

   (b) What is the region of the w plane into which the rectangular region in the z-plane bounded by $x=0, y=0, x=2, y=1$ is mapped under the transformation $w=z+(2-i)$. **(10)**

   (c) Find $\int\sqrt{1+\sqrt{1+z}}\,dz$, stating conditions under which your result is valid. **(10)**

3. (a) Determine the poles of $\dfrac{3+4z}{z(z+1)^2(z-3)}$, over $C:|z|=4$ and residue at each pole.

   Hence find $\int_C \dfrac{3+4z}{z(z+1)^2(z-3)}dz$, over $C:|z|=4$ **(15)**

   (b) Compute $\int_C \dfrac{7-2z}{(z+1)(z-1)(z-3)}dz$ over $C:|z|=\dfrac{5}{2}$ with the help of Cauchy's Integral Formula. **(10)**

   (c) Expand $f(z)=\dfrac{z}{(z-1)(2-z)}$ in a Laurent Series valid for $0<|z-2|<1$. **(10)**

4. Compute the following using Contour integration:

   (a) $\int_0^{2\pi}\dfrac{\cos 2\theta}{5+4\cos\theta}d\theta$ **(17)**

   (b) $\int_{-\infty}^{\infty}\dfrac{dx}{1+x^6}$ **(18)**

Contd P/2

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) Draw a suitable diagram for the following distribution. Read the median from the graph and verify the result by formula. How many workers earned monthly wages between 2000 and 2170 Tk.? **(18)**

Monthly wages	No. of workers	Monthly wages	Workers	Monthly wages	Workers
1900-1950	6	2050-2100	30	2200-2400	15
1950-2000	10	2100-2150	16		
2000-2050	22	2150-2200	12		

(b) Answer the followings: **(17)**

(i) The sample consists of 34 observations recorded correct to the nearest integer, ranging in value from 201 to 337. If it is decided to use seven classes of width 20 integers and to begin the first class at 199.5, find the class limits and class marks of the seven classes.

(ii) Two distributions have same values of mean, SD and Skewness. Find Kurtosis of them. Comment on your result.

6. (a) Suppose that 500 lottery tickets are sold. Among them, 200 tickets pay off at least the cost of the ticket. Now suppose that you buy 5 tickets. Find the probability that you will win back at least the cost of 3 tickets. **(10)**

(b) The probabilities are 0.4, 0.2, 0.3, and 0.1, respectively, that a delegate to a certain convention arrived by air, bus, Automobile, or train. What is the probability that among 9 delegates randomly selected at this convention, 3 arrived by air, 3 arrived by bus, 1 arrived by automobile, and 2 arrived by train? **(10)**

(c) An overseas shipment of 5 automobile contains 2 that have slight paint blemishes. If an agency receives 3, then find probability distribution and F(x) of the random variable X representing no. of automobiles purchased by the agency with paint blemishes. **(15)**

7. (a) It is important that scientific researchers in the area of forest products be able to study correlation among the anatomy and mechanical properties of trees. For the study Quantitative Anatomical Characteristics of Plantation Grown Loblolly Pine (Pinus Taeda L.) and Cottonwood (Populus deltoids Bart. Ex Marsh.) and their relationships to mechanical properties, conducted by the Department of Forestry and Forest Products at Virginia Tech, 15 loblolly pines are randomly selected for investigation. The resulting data on the specific gravity and the modulus of rupture is shown below: **(20)**

Contd P/3

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MATH 245/CSE

Contd.... for Q. No. 7(a)

Specific Gravity, x (g/cm^3)	Modulus of Rupture, y (kPa)
0.414	29,186
0.383	29,266
0.399	26,215
0.402	30,162
0.442	38,867
0.422	37,831
0.466	44,576
0.500	46,097
0.514	59,698
0.530	67,705
0.569	66,088
0.558	78,486
0.577	89,869
0.572	77,369
0.548	67,095

(i) Find both the regression coefficients.

(ii) Compute and interpret the sample correlation coefficient.

(b) In an industrial process the diameter of a ball bearing is an important component part. The buyer sets specifications on the diameter to be 3.0 ± 0.01 cm. The implication is that no part falling outside these specifications will be accepted. It is known that in the process the diameter of a ball bearing has a normal distribution with mean 3.0 and standard deviation 0.005. On the average, how many manufactured ball bearings will be scrapped? (Table 1 is attached.) **(15)**

8. (a) Ten individuals are chosen at random from a population and their heights are found to be inches 63, 63, 64, 65, 66, 69, 69, 70, 70, 71. Discuss the suggestion that the mean height of universe is 65. Assume that for 9 degrees of freedom t at 5% level of significance is 2.262. **(17)**

(b) A merchandising company wishes to test whether its 3 salesman A, B and C tend to make sales of the same size or whether they differ selling ability as measures by the average size of their sales. During the last week, out of 14 sales, A made 5, B made 4 and C made 5 calls. The following are the weekly sales (in Tk thousand) record of 3 salesman: **(18)**

A	B	C
300	600	700
400	300	300
300	300	400
500	400	600
0	—	500

Test whether the three salesman's average sales differ in size. (Given that $F_{(2,11)}$ at 5% level of significance = 3.98)

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Table 1: Standard normal distribution values ($z \geq 0$)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91308	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983

Table 1 for Question 7(b)

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L-2/T-2/CSE Date: ~~01/02/2025~~

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-2 B. Sc. Engineering Examinations 2022-2023

Sub: **MATH 243** (Probability and ~~Statics~~) Statistics

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks.

Symbols used have their usual meaning.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) One bag contains 4 white balls and 3 black balls, and a second bag contains 3 white balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is probability that a ball now drawn from the second bag is black? **(8)**

(b) In a certain assembly plant, three machines, B_1, B_2, and B_3, make 30%, 45%, and 25% respectively, of the products. It is known from the past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected and it is defective. What is the probability that this product is made by machine **(15)**

$B_i, i = 1, 2, 3$?

(c) The time to failure in hours of an important piece of electronic equipment used in a manufactured DVD player has the density function **(12)**

$$f(x) = \begin{cases} \frac{1}{2000}\exp\left(-\frac{x}{2000}\right), & \text{when } x \geq 0 \\ 0, & \text{when } x < 0 \end{cases}$$

i. Find cumulative distribution function for $f(x)$.

ii. Determine the probability that the component (and thus the DVD player) lasts more than 1000 hours before the component needs to be replaced.

iii. Determine the probability that the component fails before 2000 hours.

2. (a) The joint density for the random variables (X, Y), where X is the unit temperature change and Y is the proportion of spectrum shift that a certain atomic particle produces, **(15)**

is $f(x,y) = \begin{cases} 10xy^2, & 0 < x < y < 1 \\ 0, & \text{elsewhere} \end{cases}$

i. Find marginal densities g(x), $h(y)$, and conditional density $f(y|x)$.

ii. Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.

Contd P/2

MATH 243/CSE

Contd ... Q. No. 2

(b) The length of time, in minutes, for an airplane to obtain clearance for takeoff at a certain airport is a random variable $Y = 3X - 2$, where X has the density function **(10)**

$$f(x) = \begin{cases} \frac{1}{4}\exp\left(-\frac{x}{4}\right), & x > 0 \\ 0, \text{ elsewhere} \end{cases}$$

Find the mean and variance of the random variable Y.

(c) Compute $p(\mu - 2\sigma < X < \mu + 2\sigma)$, where X has the density function **(10)**

$$f(x) = \begin{cases} 6x(1-x), & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

and compare with the result given in Chebyshev's theorem.

3. (a) Briefly discuss about Bernoulli and Poisson process. Derive Poisson distribution for Poisson random variable. **(10)**

(b) Calculate the variance for Poisson random variable. Now, suppose that, on average, 1 person in 1000 makes a numerical error in preparing his or her income tax return. If 10,000 returns are selected at random and examined, find the probability that 6, 7, or 8 of them contain an error. **(12)**

(c) A lawyer commutes daily form his suburban home to his midtown office. The average time for a one-way trip is 24 minutes, with a standard deviation of 3.8 minutes. Assume the distribution of trip times to be normally distributed. What is the probability that a trip will take at least half hour? **(13)**

i. If the office opens at 9:00 am and the lawyer leaves his house at 8:45 am daily, what percentage of the time is he late for work?

ii. Find the length of time above which we find the slowest 15% of the trips

iii. Find the probability that 2 of the next 3 trips will take at least half hour.

(Necessary table is attached)

4 (a) Write down some properties of normal curve. Find the mean, mode, and standard deviation of normal random variable. Hence comment on the mean and variance of standard normal random variable. **(10)**

(b) Distinguish clearly between correlation and regression analysis. An examination of eight applicants for a post was taken by a firm. From the marks obtained by the applications in the Bangla and English papers, compute the rank correlation coefficients and comment on your findings: **(12)**

Applicant	A	B	C	D	E	F	G	H
Marks in Bangla	15	20	28	12	40	60	20	80
Marks in English	40	30	50	30	20	10	30	60

Contd P/3

MATH 243/CSE

Contd ... Q. No. 4

(c) From the data given below find out- **(13)**

i The two linear regression equations,

ii. Most likely age of husband when the wife's age is 30.

iii. Most likely age of wife when the husband's age is 25.

Ages of husbands	22	23	23	24	26	27	27	28	30	30
Ages of wives	18	20	21	20	21	22	23	24	25	26

Also, comment on Pearson's correlation coefficient between the above variables.

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) What do you mean by statistical data? How are they produced? Why do you need statistical data at all? What are the usual sources of statistical data in Bangladesh? **(10)**

(b) A random sample was taken of the thickness of insulation in transformer windings, and the following thickness (in millimeters) were recorded: **(15)**

18	21	22	29	25	31	37	38	41	39
44	48	54	56	56	57	47	38	35	36
29	37	32	42	43	40	48	36	37	37
36	38	40	41	44	39	38	34	24	32
39	44	42	30	37	30	42	37	34	37
32	24	42	36	49	39	23	34	36	40

(i) Represent a frequency table for the data using Sturges' rule $K = 1+3.322\log N$ where K is the number of classes and N is the total frequency.

(ii) Draw a frequency histogram and a cumulative frequency graph.

(c) A purchasing agent obtained samples of 60-watt bulbs from two companies. He/She had the sample tested in his own laboratory for length of life with the following results: **(10)**

	Samples from	
Length of life (in hours)	Company A	Company B
1700 and under 1900	10	3
1900 and under 2100	16	40
2100 and under 2300	20	12
2300 and under 2500	8	3
2500 and under 2700	6	2

If prices of both types are the same, select which company's bulbs you will prefer to collect. Justify your preference.

Contd P/4

MATH 243/CSE

6. (a) Define and illustrate with examples: **(10)**

(i) Standard deviation, (ii) Variance, (iii) Skewness, (iv) Kurtosis and (v) Moments.

(b) The following data represent the length in centimeters (cm) of 12 pencils: **(10)**

17.2	16.4	18.6	17.4	17.5	19.0
17.4	16.8	19.0	17.8	18.8	18.7

Use a stem and leaf plot to display the data. Hence find the median, mode and range of pencil length?

(c) An analysis of companies resulted in the following distribution: **(15)**

Profit (Lakhs)	10-20	20-30	30-40	40-50	50-60
No of Companies	18	20	30	22	10

Calculate the first four moments about assumed mean. Convert the result into moments about the mean. Compute the value of γ_1 and γ_2 and comment on the result.

7. (a) Differentiate the following pair of concepts: **(10)**

(i) Statistic and parameter, (ii) Critical region and acceptable region, (iii) Exact and inexact hypothesis, (iv) Unidirectional and non-directional test, (v) Level of significance and p-value.

(b) The mean and standard deviation of GPA scores obtained from a random sample of 40 student of a college were 2.8 and 0.35 respectively. Can we conclude that the sample has come from the entire group of students, which has a mean score of 2.4? Compute 99 percent confidence interval for the mean score in the population. (Given that at 1% level of significance; $z = \pm 2.58$). **(12)**

(c) The average waiting time in a bank counter to each a cheque for all customers has been 50 minutes. A new service-providing procedure using modern computer facilities is being tried. If a random sample of 12 customers had an average waiting time for service is 42 minutes with a standard deviation of 11.9 minutes under the new system, test the hypothesis that the population mean is now less than 50, using a significance level of 5 percent assuming the waiting time to be normally distributed. (Given that at $v = 11$, $t_{0.05} = 1.796$). **(13)**

8. (a) A certain drug was administered to 500 people out of a total 800 included in the sample to test its efficacy against typhoid. The results are given below: **(10)**

	Typhoid	No. Typhoid	Total
Drug	200	300	500
No. Drug	280	20	300
Total	480	320	800

Based on these data, can it be concluded that the drug is effective in preventing typhoid using 5% level of significance. (Given that at $v = 1$, $\chi^2_{0.05} = 3.84$).

Contd P/5

MATH 243/CSE

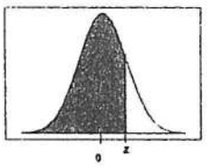
Contd ... Q. No. 8

(b) As head of a department of a consumer's research organization, you have the responsibility for testing and comparing lifetimes of four brands of electric bulbs. Suppose you test the lifetime of three electric bulbs of each of the four brands. The data is shown below, each entry representing the lifetime of an electric bulb, measured in hundreds of hours: Can we infer that the mean lifetime of the four brands of electric bulbs are equal with 5% and 1% level of significance? (Necessary charts are attached). **(10)**

Brand			
Brand A	Brand B	Brand C	Brand D
20	25	24	23
19	23	20	20
21	21	22	20

(c) Seeds of four different types of corn are planted in five blocks. Each block is divided into four plots, which are then randomly assigned to the four types of corn. Determine at the 0.05 significance level whether the yields in bushels per acre, as shown in the following Table, vary significantly with differences in (i) the soil (i.e., the five blocks) and (ii) the type of corn. (Necessary charts are attached). **(15)**

	Types of corn			
	I	II	III	IV
Block A	12	15	10	14
Block B	15	19	12	11
Block C	14	18	15	12
Block D	11	16	12	16
Block E	16	17	11	14

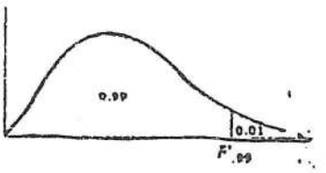


Cumulative probabilities for POSITIVE z-values are in the following table:

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

99th Percentile Values for the *F* Distribution

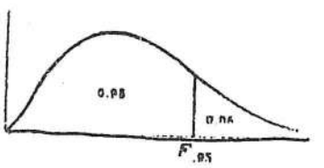
(ν_1 degrees of freedom in numerator)
(ν_2 degrees of freedom in denominator)



ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	4052	5000	5403	5625	5764	5859	5928	5981	6023	6056	6106	6157	6209	6235	6261	6287	6313	6339	6366
2	98.5	99.0	99.2	99.2	99.3	99.3	99.4	99.4	99.4	99.4	99.4	99.4	99.4	99.5	99.5	99.5	99.5	99.5	99.5
3	34.1	30.8	29.5	28.7	28.2	27.9	27.7	27.5	27.3	27.2	27.1	26.9	26.7	26.6	26.5	26.4	26.3	26.2	26.1
4	21.2	18.0	16.7	16.0	15.5	15.2	15.0	14.8	14.7	14.5	14.4	14.2	14.0	13.9	13.8	13.7	13.7	13.6	13.6
5	16.3	13.3	12.1	11.4	11.0	10.7	10.5	10.3	10.2	10.1	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
6	13.7	10.9	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
7	12.2	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
8	11.3	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86
9	10.6	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31
10	10.0	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17
14	8.86	6.51	5.56	5.04	4.70	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75
17	8.40	6.11	5.19	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.36
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	7.77	5.57	4.68	4.18	3.86	3.63	3.46	3.32	3.22	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	2.96	2.82	2.66	2.58	2.50	2.42	2.33	2.23	2.13
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.90	2.75	2.60	2.52	2.44	2.35	2.26	2.17	2.06
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53	1.38
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

Chart-1 for Q.8

95th Percentile Values
for the F Distribution
(ν_1 degrees of freedom in numerator)
(ν_2 degrees of freedom in denominator)



ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	161	200	216	225	230	234	237	239	241	242	244	246	248	249	250	251	252	253	254
2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5	19.5	19.5	19.5	19.5
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	290	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Chart-1 for Q-8